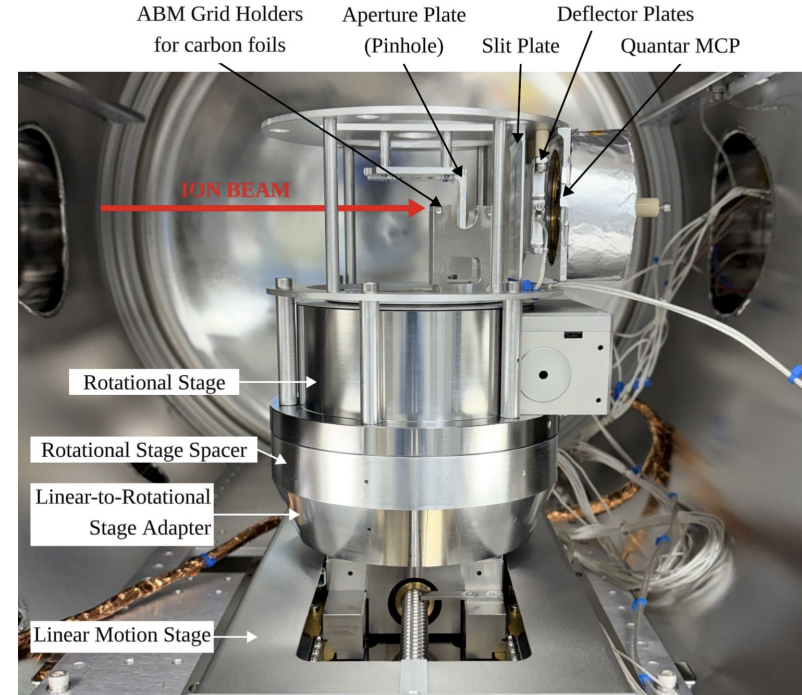
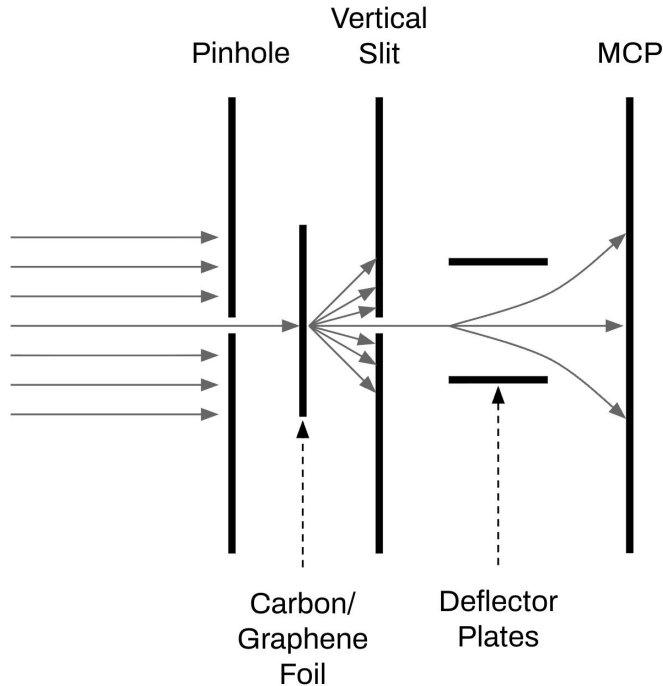


# **First Light Data Graphene & $0.6 \mu\text{g}/\text{cm}^2$ Carbon Foils GEMINI Instrument**

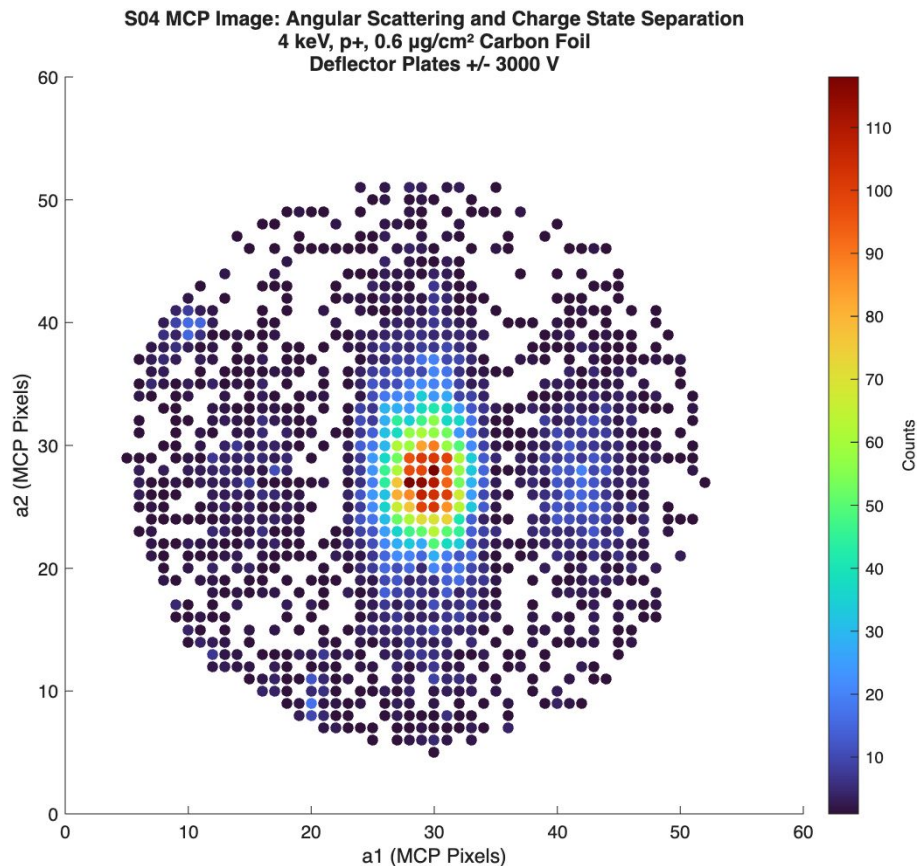
William Li - Junior Independent Work - Spring 2026  
Space Physics at Princeton

# Recap of Instrument Design

- Oppositely charged Deflector Plates separate particles by charge state horizontally on the MCP
- Angular scattering measured vertically on the MCP

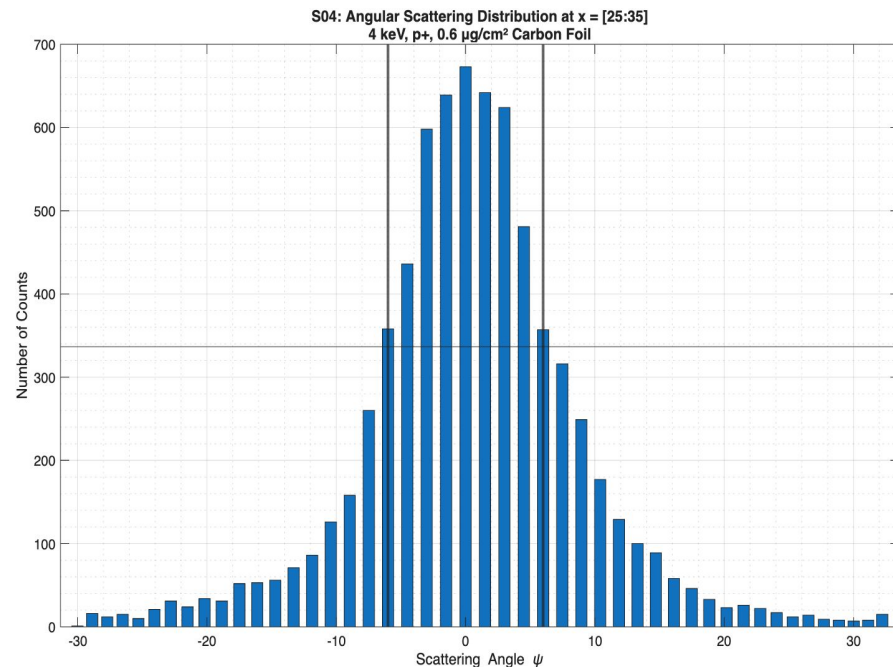
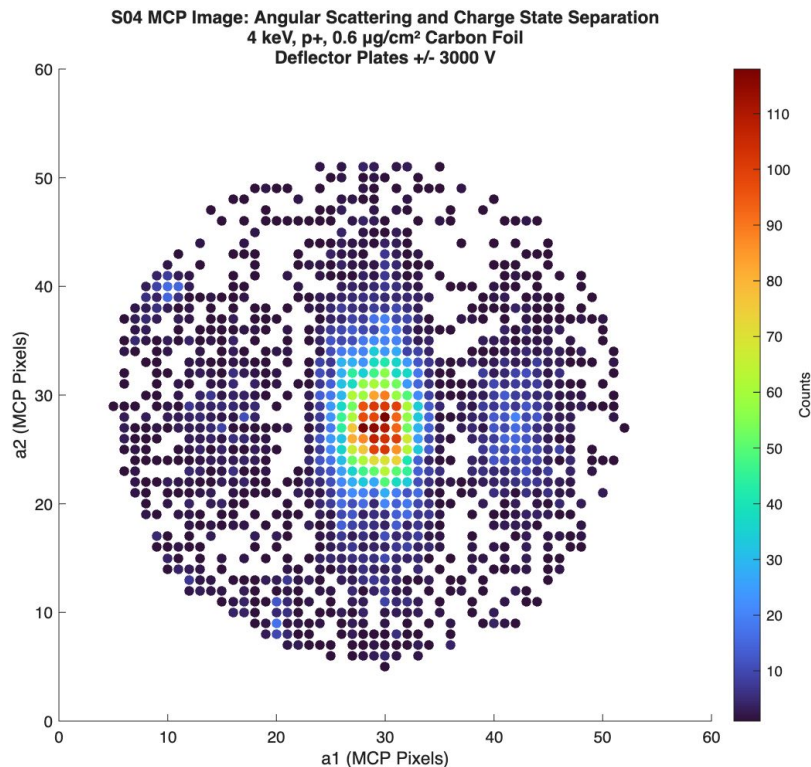


# 4 keV, $\text{H}^+$ , $0.6 \mu\text{g}/\text{cm}^2$ Carbon Foil, 3000 V



# 4 keV, H<sup>+</sup>, 0.6 μg/cm<sup>2</sup> Carbon, 3000 V, Scattering

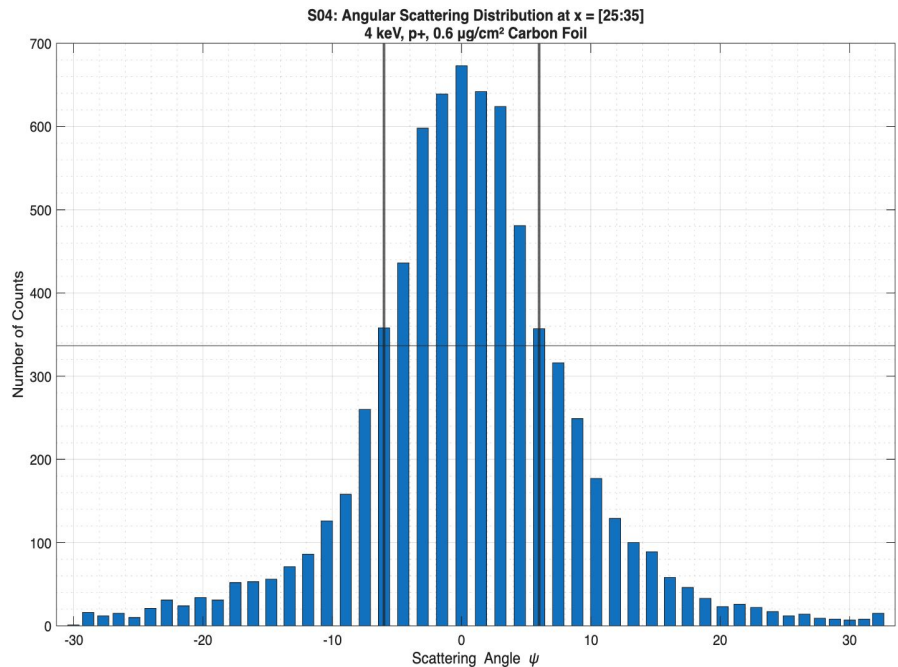
- Vertical cross-section from  $x = 25$  to  $x = 35$



Scattering Half-Angle  $\psi_{1/2} = 6^\circ$



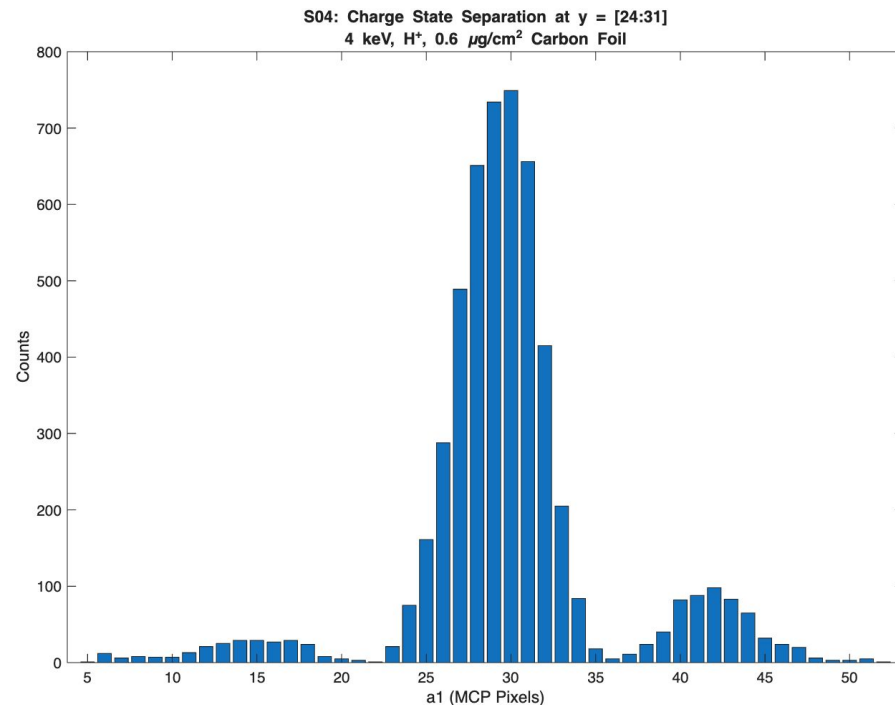
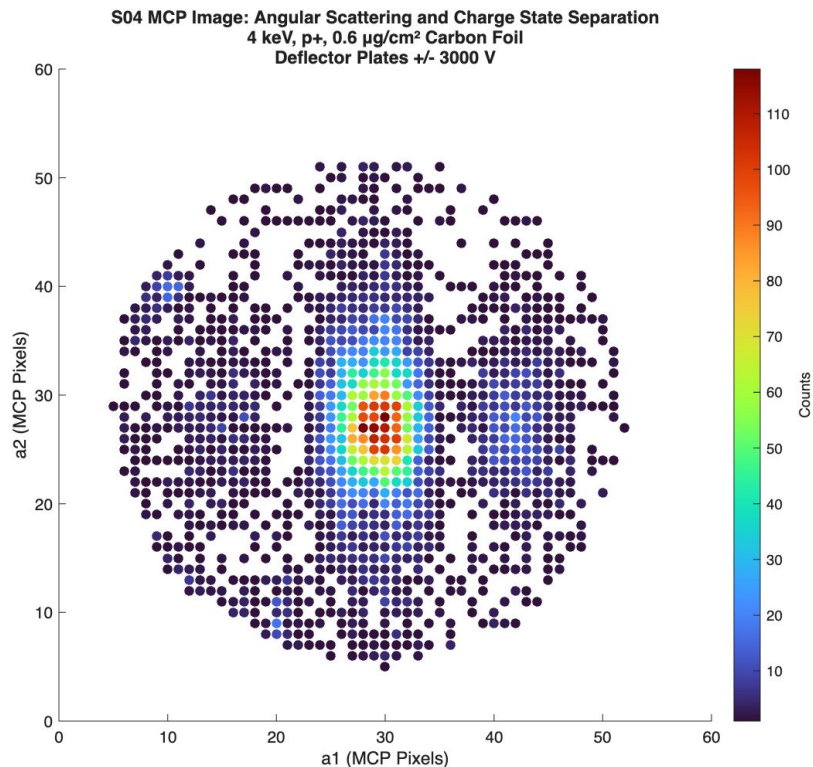
spacephysics.  
princeton.edu



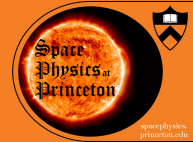
5

# 4 keV, H<sup>+</sup>, 0.6 μg/cm<sup>2</sup> Carbon, 3000 V, Charge State

- Horizontal cross-section from y = 24 to y = 31



# 4 keV, $H^+$ , Charge State Comparison to Allegrini et al. 2014



## Allegrini et al. 2014

- ~10% Positive
- ~84% Neutral
- ~6% Negative

## My Data

- ~11% Positive
- ~85% Neutral
- ~3% Negative

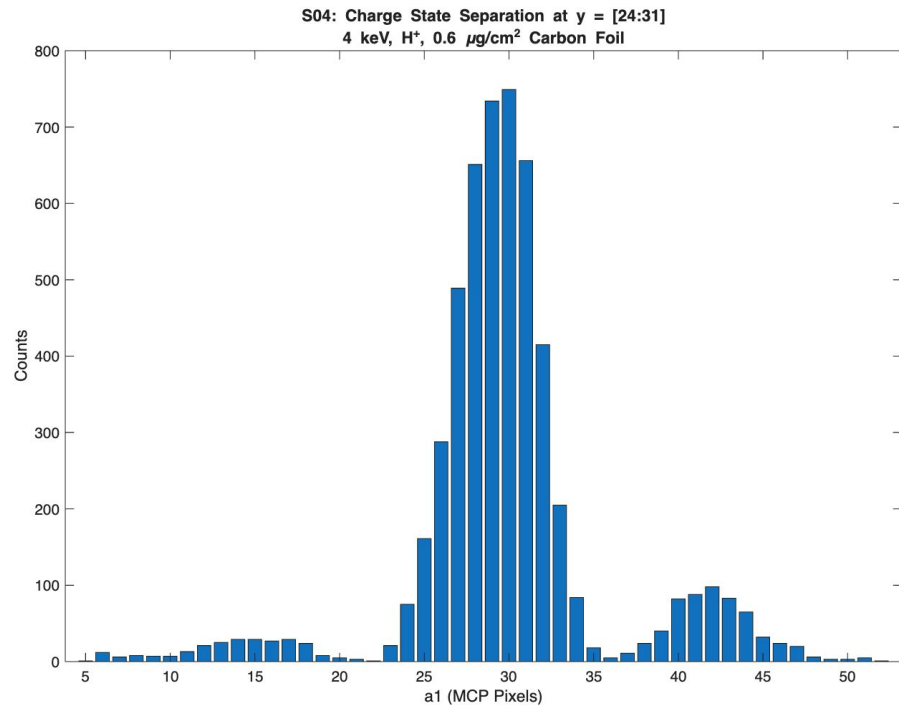
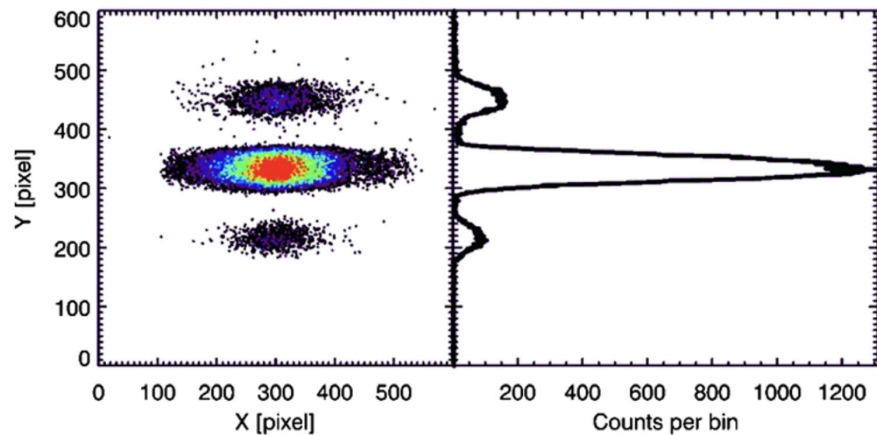
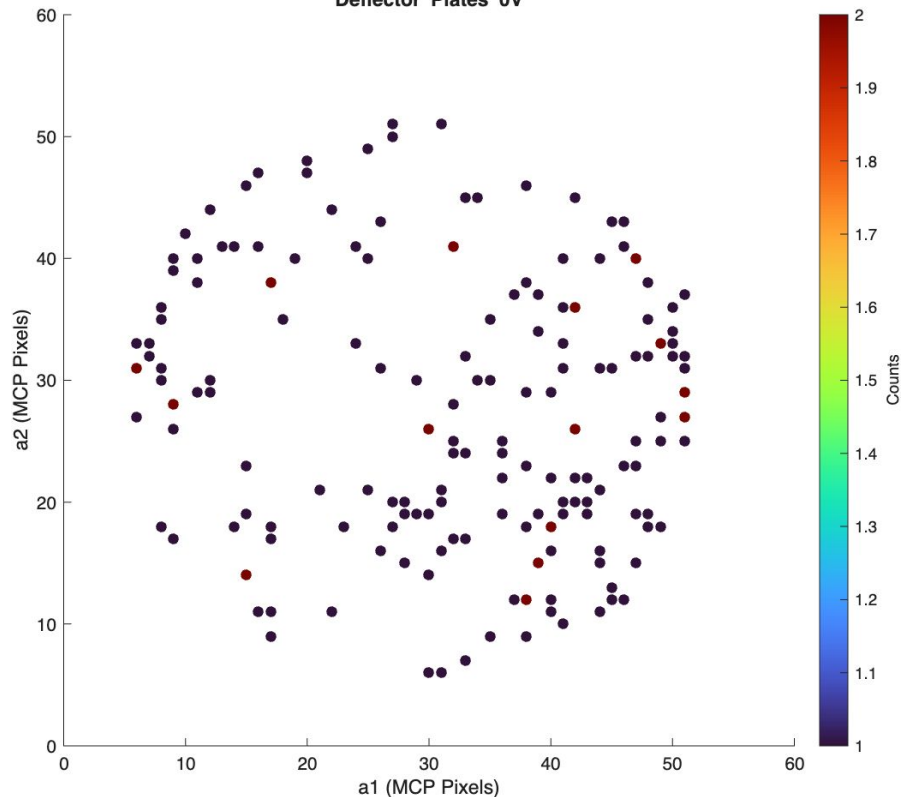


Figure from Allegrini et al. 2014

# Graphene PMMA Layer Blocks up to 16 keV H<sup>+</sup> Beam?

FL12: 1 keV, H<sup>+</sup>, Graphene  
Deflector Plates 0V



G04: 16 keV, H<sup>+</sup>, Graphene  
Deflector Plates 0V

